Intensive Care Management MCQ – Chapter 40

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 9: Cardiorespiratory Care (Chapter 40)

50 Questions • 30 Minutes

Instructions:
- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. The first intensive care unit (ICU) was started in which year?

A. 1945
B. 1950
C. 1953
D. 1960

Ans: C

Q2. What percentage of hospital beds should be allocated to the ICU?

A. 5%
B. 10%
C. 15%
D. 20%

Ans: B

Q3. Acute respiratory failure is considered to be present when pO■ is less than:

A. 40 mm Hg
B. 50 mm Hg
C. 60 mm Hg
D. 70 mm Hg

Ans: C

Q4. Elevated pCO■ in the presence of normal pH constitutes:

A. Acute respiratory failure
B. Chronic respiratory failure
C. Combined respiratory failure
D. Metabolic acidosis

Ans: B

Q5. Type II respiratory failure is characterized by all of the following EXCEPT:

A. Increased pCO■ (>50 mm Hg)
B. pO■ is usually normal
C. Increased pAO■–paO■ gradient
D. Normal pAO■–paO■ gradient

Ans: C

Q6. Which of the following is a cause of Type II (hypercapnic ventilatory) respiratory failure?

A. Pulmonary embolism
B. Pulmonary edema
C. Guillain-Barré syndrome
D. Interstitial lung disease

Ans: C

Q7. Oxygen therapy is helpful in improving oxygen saturation in all types of hypoxia EXCEPT:

A. Hypoxic hypoxia
B. Anemic hypoxia
C. Stagnant hypoxia
D. Histotoxic hypoxia

Ans: D

Q8. Ideally, inspired oxygen concentration (FiO■) should not exceed more than what percentage to avoid oxygen toxicity?

- **A.** 30%
- **B.** 40%
- **C.** 50%
- **D.** 60%

Ans: C

Q9. Which of the following is an indication for mechanical ventilation based on pulmonary function?

- **A.** Vital capacity >15 mL/kg
- **B.** Respiratory rate >35/minute
- **C.** Dead space volume (VD/VT) <0.6
- **D.** Tidal volume >5 mL/kg

Ans: B

Q10. On the basis of blood gas analysis, mechanical ventilation is indicated when pO■/FiO■ is less than:

- **A.** 150 mm Hg
- **B.** 200 mm Hg
- **C.** 250 mm Hg
- **D.** 300 mm Hg

Ans: C

Q11. The chief advantage of volume preset (volume controlled) ventilation is:

- **A.** Decreased chances of barotrauma
- **B.** Assurance of delivery of preset tidal volume
- **C.** Maintenance of preset airway pressure
- **D.** Better patient comfort

Ans: B

Q12. The usual initial tidal volume setting on a ventilator is:

- **A.** 4–5 mL/kg
- **B.** 6–8 mL/kg
- **C.** 10–12 mL/kg
- **D.** 12–15 mL/kg

Ans: B

Q13. The normal inspiratory to expiratory (I:E) ratio set on a ventilator is:

- **A.** 1:1
- **B.** 1:2
- **C.** 2:1
- **D.** 1:3

Ans: B

Q14. The major disadvantage of controlled-mode ventilation (CMV) is:

- **A.** Hyperventilation
- **B.** Increased work of breathing
- **C.** Intrathoracic pressure always remains positive, decreasing venous return and cardiac output
- **D.** Cannot deliver adequate tidal volume

Ans: C

Question	Answer
Q15. In Assist-Control (A/C) ventilation, if the preset rate is 10 breaths/min and the patient's spontaneous rate is 12, the ventilator will: **A.** Deliver 10 breaths in controlled mode **B.** Assist 10 breaths and deliver 2 in control mode **C.** Assist all 12 breaths **D.** Deliver 12 breaths in controlled mode	**Ans: C**
Q16. Which of the following is an advantage of SIMV over CMV? **A.** Less work of breathing **B.** Less hypotension due to spontaneous breaths allowing good venous return **C.** Better control of tidal volume **D.** Less hyperventilation	**Ans: B**
Q17. In Inverse Ratio Ventilation (IRV), the I:E ratio is reversed to: **A.** 1:1 **B.** 1:3 **C.** 2:1 **D.** 3:1	**Ans: C**
Q18. In Pressure Support Ventilation (PSV), the ventilator cycles to expiration when the predetermined flow falls below: **A.** 10% **B.** 15% **C.** 20% **D.** 25%	**Ans: D**
Q19. The most commonly used dual mode of ventilation in India is: **A.** Adaptive support ventilation (ASV) **B.** Pressure-regulated volume control (PRVC) **C.** Volume-assured pressure support (VAPS) **D.** Automode ventilation	**Ans: B**
Q20. High-frequency jet ventilation operates at a frequency of: **A.** 60–120 cycles/min **B.** 100–300 cycles/min **C.** 300–600 cycles/min **D.** 600–3000 cycles/min	**Ans: B**
Q21. Which is the most commonly used mode for high-frequency ventilation? **A.** High-frequency positive-pressure ventilation **B.** High-frequency jet ventilation **C.** High-frequency oscillation **D.** High-frequency percussive ventilation	**Ans: B**

Q22. The ideal PEEP is defined as the PEEP which maintains oxygen saturation >90% without decreasing the cardiac output significantly. The usual range is:

A. 2–5 cm H■O
B. 5–12 cm H■O
C. 12–20 cm H■O
D. 20–25 cm H■O

Ans: B

Q23. CPAP (Continuous Positive Airway Pressure) is essentially:

A. A mode of mechanical ventilation
B. PEEP given to a nonintubated patient
C. Pressure support ventilation through a mask
D. Bilevel positive airway pressure

Ans: B

Q24. In BIPAP, the typical inspiratory positive airway pressure (IPAP) setting is:

A. 2–5 cm H■O
B. 5–10 cm H■O
C. 8–20 cm H■O
D. 20–30 cm H■O

Ans: C

Q25. What is the most common indication of NIPPV in the acute setting?

A. Sleep apnea syndrome
B. Acute exacerbation of COPD
C. Cardiogenic pulmonary edema
D. Pneumothorax

Ans: B

Q26. What is the major problem during NIPPV?

A. Aspiration
B. Claustrophobia
C. Leakage
D. Skin breakdown

Ans: C

Q27. For weaning from ventilator, the Rapid Shallow Breathing Index (RSBI) should be:

A. <50
B. <80
C. <100
D. <150

Ans: C

Q28. Weaning from ventilator is possible in all modes of ventilation EXCEPT:

A. Assist-Control mode
B. SIMV mode
C. Controlled-mode ventilation
D. Pressure Support Ventilation

Ans: C

Q29. **The incidence of pulmonary barotrauma during mechanical ventilation is:** **A.** 1–3% **B.** 5–7% **C.** 7–10% **D.** 15–20%	**Ans: C**

Q30. **The most common source of nosocomial infections in ICU is:** **A.** Pulmonary infections **B.** Urinary tract infections **C.** Bloodstream infections **D.** Wound infections	**Ans: B**

Q31. **Ventilator-associated pneumonia (VAP) is defined as pneumonia developing in patients on ventilators for more than:** **A.** 24 hours **B.** 48 hours **C.** 72 hours **D.** 96 hours	**Ans: B**

Q32. **The simplest and most cost-effective method to decrease ventilator-associated pneumonia is:** **A.** Use of subglottic suctioning endotracheal tubes **B.** Oral hygiene with chlorhexidine **C.** Positioning head of bed at 30 degrees **D.** Early tracheostomy	**Ans: C**

Q33. **Shock is classified into how many major types?** **A.** Two **B.** Three **C.** Four **D.** Five	**Ans: C**

Q34. **In cardiogenic shock, the pulmonary capillary wedge pressure (PCWP) in left ventricular failure is typically:** **A.** >15 mm Hg **B.** >18 mm Hg **C.** >25 mm Hg **D.** >30 mm Hg	**Ans: C**

Q35. **The hallmark hemodynamic finding in distributive shock is:** **A.** Increased CVP **B.** Increased PCWP **C.** Decreased SVR **D.** Decreased cardiac output	**Ans: C**

Q36. In hemorrhagic shock, the current guidelines recommend maintaining systolic blood pressure (SBP) between:

- **A.** 60–80 mm Hg
- **B.** 80–100 mm Hg
- **C.** 100–120 mm Hg
- **D.** 120–140 mm Hg

Ans: B

Q37. The concept of combining hypotensive resuscitation and hemostatic resuscitation is called:

- **A.** Goal-directed therapy
- **B.** Balanced resuscitation
- **C.** Damage control resuscitation
- **D.** Permissive hypotension

Ans: B

Q38. Which parameter is considered the most reliable to assess fluid status and titrate fluid therapy?

- **A.** Central venous pressure (CVP)
- **B.** Pulse pressure variation
- **C.** Stroke volume variation
- **D.** Pulmonary artery occlusion pressure

Ans: C

Q39. The most preferred crystalloid for fluid resuscitation in shock is:

- **A.** Normal saline
- **B.** Ringer's lactate
- **C.** 5% dextrose
- **D.** Dextrose-normal saline

Ans: B

Q40. The vasopressor of choice for septicemic shock is:

- **A.** Phenylephrine
- **B.** Ephedrine
- **C.** Norepinephrine (noradrenaline)
- **D.** Vasopressin

Ans: C

Q41. Steroids are contraindicated in which type of shock?

- **A.** Septicemic shock
- **B.** Hypovolemic shock
- **C.** Cardiogenic shock
- **D.** Neurogenic shock

Ans: C

Q42. The vein of choice for parenteral nutrition is:

- **A.** Internal jugular vein
- **B.** Subclavian vein
- **C.** Femoral vein
- **D.** Antecubital vein

Ans: B

Q43. **Out of total energy requirement, what percentage should be provided by carbohydrates?**

A. 40–50%
B. 50–60%
C. 60–70%
D. 70–80%

Ans: C

Q44. **Sepsis is the leading cause of death in ICU. The most common cause of death in septicemia is:**

A. Cardiogenic shock
B. ARDS and renal failure
C. Disseminated intravascular coagulation
D. Hepatic failure

Ans: B

Q45. **Pulmonary edema develops when pulmonary artery occlusion pressure (left atrial pressure) rises to more than:**

A. 15 mm Hg
B. 18 mm Hg
C. 25 mm Hg
D. 35 mm Hg

Ans: C

Q46. **According to the Berlin definition, severe ARDS is defined as PaO■/FiO■:**

A. ≤300 mm Hg with PEEP ≥5 cm H■O
B. ≤200 mm Hg with PEEP ≥5 cm H■O
C. ≤100 mm Hg with PEEP ≥5 cm H■O
D. ≤50 mm Hg with PEEP ≥10 cm H■O

Ans: C

Q47. **The most common cause of ARDS is:**

A. Polytrauma
B. Pancreatitis
C. Sepsis
D. Aspiration of gastric contents

Ans: C

Q48. **Carbon monoxide poisoning is usually defined as carboxyhemoglobin greater than what percentage in blood?**

A. 10%
B. 15%
C. 20%
D. 30%

Ans: C

Q49. **The normal anion gap is:**

A. 4–8 mM/L
B. 8–16 mM/L
C. 16–24 mM/L
D. 24–32 mM/L

Ans: B

Q50. The most common cause of metabolic acidosis in anesthesia is: **A.** Renal failure **B.** Diabetic ketoacidosis **C.** Lactic acidosis due to tissue hypoxia **D.** Diarrhea with bicarbonate loss	**Ans: C**

Answer Key & Explanations

Q1. Answer: C — 1953

According to the textbook, the first intensive care unit (ICU) was started in 1953.

Topic: ICU Setup | Ref: Ch 40, Section: Intensive Care Unit, Page 277

Q2. Answer: B — 10%

Total number of beds in ICU should be 10% of hospital beds, with each bed having 20 m^2 floor area.

Topic: ICU Setup | Ref: Ch 40, Section: Intensive Care Unit, Page 277

Q3. Answer: C — 60 mm Hg

Acute respiratory failure is considered to be present if pO■ is less than 60 mm Hg, pCO■ over 50 mm Hg and pH less than 7.30.

Topic: Acute Respiratory Failure | Ref: Ch 40, Section: Acute Respiratory Failure – Definition, Page 277

Q4. Answer: B — Chronic respiratory failure

Elevated pCO■ in presence of normal pH constitutes chronic respiratory failure, as pH is compensated by renal tubular reabsorption of bicarbonate.

Topic: Acute Respiratory Failure | Ref: Ch 40, Section: Acute Respiratory Failure – Definition, Page 277

Q5. Answer: C — Increased pAO■–paO■ gradient

In Type II (hypercapnic ventilatory failure), pCO■ is >50 mm Hg, pO■ is usually normal, and pAO■–paO■ gradient is normal. Increased pAO■–paO■ gradient is a feature of Type I failure.

Topic: Classification of Respiratory Failure | Ref: Ch 40, Section: Classification, Page 277–278

Q6. Answer: C — Guillain-Barré syndrome

Guillain-Barré syndrome is a disorder affecting signal transmission to respiratory muscles and is a cause of Type II failure. The other options are causes of Type I failure.

Topic: Classification of Respiratory Failure | Ref: Ch 40, Section: Type II Failure, Page 277–278

Q7. Answer: D — Histotoxic hypoxia

Oxygen therapy is helpful in improving oxygen saturation in all types of hypoxia except histotoxic hypoxia. Maximum benefit is seen in hypoxic hypoxia.

Topic: Supplemental Oxygen | Ref: Ch 40, Section: Supplemental Oxygen, Page 278

Q8. Answer: C — 50%

Ideally, inspired oxygen concentration (FiO■) should not exceed more than 50%, otherwise oxygen toxicity can occur.

Topic: Supplemental Oxygen | Ref: Ch 40, Section: Supplemental Oxygen, Page 278

Q9. Answer: B — Respiratory rate >35/minute

Indications for mechanical ventilation based on pulmonary functions include respiratory rate >35/min, vital capacity <15 mL/kg, VD/VT >0.6, and tidal volume <5 mL/kg.

Topic: Mechanical Ventilation | Ref: Ch 40, Section: Mechanical Ventilation – Indications, Page 278

Q10. Answer: C — 250 mm Hg

Mechanical ventilation is indicated when pO_2/FiO_2 <250 mm Hg (normal >400).

Topic: Mechanical Ventilation | Ref: Ch 40, Section: Mechanical Ventilation – Indications, Page 278

Q11. Answer: B — Assurance of delivery of preset tidal volume

The chief advantage of volume preset breath is assurance of delivery of preset tidal volume and hence decreased chances of hypoventilation; however, the chief disadvantage is increased chances of barotrauma.

Topic: Ventilators | Ref: Ch 40, Section: Ventilators, Page 279

Q12. Answer: B — 6–8 mL/kg

The usual initial tidal volume set on ventilator is 6–8 mL/kg.

Topic: Ventilator Settings | Ref: Ch 40, Section: Initial Setting of Ventilator, Page 279

Q13. Answer: B — 1:2

The usual initial I:E ratio set on ventilator is 1:2.

Topic: Ventilator Settings | Ref: Ch 40, Section: Initial Setting of Ventilator, Page 279

Q14. Answer: C — Intrathoracic pressure always remains positive, decreasing venous return and cardiac output

The major disadvantage of CMV is that the intrathoracic pressure always remains positive, decreasing the venous return and cardiac output.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – CMV, Page 279

Q15. Answer: C — Assist all 12 breaths

In A/C mode, if the patient's spontaneous rate exceeds the set rate (e.g., 12 vs set rate of 10), the ventilator will assist all 12 breaths, working in assist mode only.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – A/C, Page 280

Q16. Answer: B — Less hypotension due to spontaneous breaths allowing good venous return

Advantages of SIMV over CMV include less hypotension (as good venous return occurs during spontaneous breaths), no need for heavy sedation/muscle relaxants, and less sense of breathlessness.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – SIMV, Page 280

Q17. Answer: C — 2:1

In IRV, the normal I:E ratio of 1:2 is reversed to 2:1. Doubling the inspiratory time doubles the gas exchange time.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – IRV, Page 280

Q18. Answer: D — 25%

In PSV, the ventilator will cycle to expiration once the predetermined flow falls below 25% (flow cycled).

Topic: Modes of Ventilation | Ref: Ch 40, Section: Modes of Ventilation – PSV, Page 280

Q19. Answer: B — Pressure-regulated volume control (PRVC)

The most commonly used dual mode in India is pressure-regulated volume control (PRVC). PRVC can be considered as the mode of choice for ventilation in current day critical care practice.

Topic: Modes of Ventilation | Ref: Ch 40, Section: Dual Mode Ventilation, Page 281

Q20. Answer: B — 100–300 cycles/min

High-frequency jet ventilation operates at a frequency of 100–300 cycles/min with gases at high pressure of 60 psi. It is the most commonly used mode for high frequency ventilation.

Topic: High-Frequency Ventilation | Ref: Ch 40, Section: High-Frequency Ventilation, Page 281

Q21. Answer: B — High-frequency jet ventilation

High-frequency jet ventilation is the most commonly used mode for high frequency ventilation, with frequency of 100–300 cycles/min.

Topic: High-Frequency Ventilation | Ref: Ch 40, Section: High-Frequency Ventilation, Page 281

Q22. Answer: B — 5–12 cm H■O

The ideal PEEP which maintains oxygen saturation >90% without decreasing the cardiac output significantly is usually 5–12 cm H■O.

Topic: PEEP | Ref: Ch 40, Section: PEEP, Page 282

Q23. Answer: B — PEEP given to a nonintubated patient

CPAP is described as a misnomer. It is just PEEP given to a nonintubated patient, i.e., the noninvasive method of giving PEEP through a tight fitting mask.

Topic: CPAP | Ref: Ch 40, Section: CPAP, Page 282

Q24. Answer: C — 8–20 cm H■O

Typical BIPAP settings are 8–20 cm H■O positive pressure during inspiration (IPAP) and 5 cm H■O during expiration (EPAP).

Topic: BIPAP | Ref: Ch 40, Section: BIPAP, Page 282

Q25. Answer: B — Acute exacerbation of COPD

In the acute setting, NIPPV is most frequently applied for acute exacerbation of COPD. In the chronic setting, it is most effectively used for sleep apnea syndrome.

Topic: NIPPV | Ref: Ch 40, Section: NIPPV – Common Indications, Page 282

Q26. Answer: C — Leakage

Leakage is a major problem during NIPPV. Significant leaks can cause hypoventilation and are best avoided by using tight fitting masks.

Topic: NIPPV | Ref: Ch 40, Section: NIPPV – Complications, Page 283

Q27. Answer: C — <100

RSBI = Respiratory rate (breaths/min) / Tidal volume (in liters). RSBI should be <100 for successful weaning.

Topic: Weaning from Ventilator | Ref: Ch 40, Section: Weaning from Ventilator, Page 283

Q28. Answer: C — Controlled-mode ventilation

It is possible to wean a patient in any mode of ventilation except control mode ventilation.

Topic: Weaning from Ventilator | Ref: Ch 40, Section: Method of Weaning, Page 283

Q29. Answer: C — 7–10%

The incidence of barotrauma is 7–10%. It may manifest as pneumothorax, pneumomediastinum, pneumopericardium, pneumoperitoneum, bronchopleural fistula and air embolism.

Topic: Complications of Mechanical Ventilation | Ref: Ch 40, Section: Complications of Mechanical Ventilation, Page 283–284

Q30. Answer: B — Urinary tract infections

The most common source of nosocomial infections in ICU are urinary tract infections accounting for 35–40% of total ICU infections, followed by pulmonary (25–30%) and bloodstream infections (20%).

Topic: ICU Infections | Ref: Ch 40, Section: Complications – Infections, Page 284

Q31. Answer: B — 48 hours

VAP is defined as pneumonia developing in patients on ventilators for >48 hours with no evidence of preintubation pneumonia. The incidence of VAP is around 20%.

Topic: VAP | Ref: Ch 40, Section: Complications – VAP, Page 284

Q32. Answer: C — Positioning head of bed at 30 degrees

Positioning the head of the bed at 30 degrees decreases gastroesophageal reflux and proves to be the simplest and most cost-effective method to decrease VAP.

Topic: VAP Prevention | Ref: Ch 40, Section: Complications – VAP, Page 284

Q33. Answer: C — Four

Shock is classified into four major types: hypovolemic, cardiogenic, distributive, and obstructive.

Topic: Shock Classification | Ref: Ch 40, Section: Shock – Classification, Page 285

Q34. Answer: C — >25 mm Hg

In cardiogenic shock, poor pump function leads to increased CVP and increased PCWP. In LVF, PCWP is >25 mm Hg.

Topic: Shock Hemodynamics | Ref: Ch 40, Section: Shock – Hemodynamic Parameters, Page 285

Q35. Answer: C — Decreased SVR

In distributive shock, poor vascular tone (decreased SVR) is the hallmark. Decreased SVR decreases the afterload, increasing the cardiac output.

Topic: Shock Hemodynamics | Ref: Ch 40, Section: Shock – Hemodynamic Parameters, Page 285

Q36. Answer: B — 80–100 mm Hg

Current guidelines favor hypotensive resuscitation, maintaining SBP between 80–100 mm Hg and MAP between 60–65 mm Hg to prevent disruption of clots and increased bleeding.

Topic: Shock Management | Ref: Ch 40, Section: Management of Shock – Early Goals, Page 286

Q37. Answer: B — Balanced resuscitation

The newer concept of hypotensive and hemostatic resuscitation is called balanced resuscitation.

Topic: Shock Management | Ref: Ch 40, Section: Management of Shock – Early Goals, Page 286

Q38. Answer: C — Stroke volume variation

Stroke volume variation is considered most reliable to assess fluid status and titrate fluid therapy; however, due to technical feasibility and cost, CVP is still most commonly used.

Topic: Shock Management – Fluids | Ref: Ch 40, Section: Treatment – Fluids, Page 286

Q39. Answer: B — Ringer's lactate

Ringer is the most preferred crystalloid as it is more physiological, except for hypochloremic alkalosis and brain injury where normal saline is preferred.

Topic: Shock Management – Fluids | Ref: Ch 40, Section: Treatment – Choice of Fluids, Page 287

Q40. Answer: C — Norepinephrine (noradrenaline)

Noradrenaline is the vasopressor of choice for septicemic shock. Phenylephrine is first-line for neurogenic shock and ephedrine is most preferred for spinal shock.

Topic: Shock Management – Drugs | Ref: Ch 40, Section: Treatment – Drugs, Page 287

Q41. Answer: C — Cardiogenic shock

Steroids are contraindicated in cardiogenic shock as they can alter the healing process of myocardium. They are effective for septicemic shock refractory to fluids and vasopressors.

Topic: Shock Management – Drugs | Ref: Ch 40, Section: Treatment – Drugs, Page 287

Q42. Answer: B — Subclavian vein

Due to large diameter (and hence less chances of thrombophlebitis due to lipid emulsions), subclavian is the vein of choice for parenteral nutrition.

Topic: Nutritional Support | Ref: Ch 40, Section: Nutritional Support – Parenteral Route, Page 288

Q43. Answer: C — 60–70%

Out of total energy required, 60–70% should be provided by carbohydrates (dextrose) and 30–40% by fats. Amino acids are given as supplement, never as a substrate for source of energy.

Topic: Nutritional Support | Ref: Ch 40, Section: Substrates for Providing Energy, Page 288

Q44. Answer: B — ARDS and renal failure

ARDS and renal failure are the most common causes of death in septicemia.

Topic: Sepsis | Ref: Ch 40, Section: Sepsis, Page 290

Q45. Answer: C — 25 mm Hg

Cardiogenic pulmonary edema develops when pulmonary artery occlusion pressure (left atrial pressure) rises to more than 25 mm Hg (or 30 cm H■O).

Topic: Pulmonary Edema | Ref: Ch 40, Section: Pulmonary Edema – Hemodynamic Type, Page 290

Q46. Answer: C — ≤100 mm Hg with PEEP ≥5 cm H■O

Berlin definition classifies ARDS as: Mild (PaO■/FiO■ ≤300), Moderate (≤200), and Severe (≤100 mm Hg), all with PEEP or CPAP ≥5 cm H■O.

Topic: ARDS | Ref: Ch 40, Section: ARDS – Berlin Definition, Page 291

Q47. Answer: C — Sepsis

Sepsis is the most common cause of ARDS.

Topic: ARDS | Ref: Ch 40, Section: Causes of ARDS, Page 291

Q48. Answer: C — 20%

Carbon monoxide poisoning is usually defined as carboxyhemoglobin >20% in blood. The affinity of CO to bind hemoglobin is 240 times that of oxygen.

Topic: Carbon Monoxide Poisoning | Ref: Ch 40, Section: Carbon Monoxide Poisoning, Page 292

Q49. Answer: B — 8–16 mM/L

Normal anion gap = 8–16 mM/L (average 12 mM/L). Anion gap = Sodium concentration – (chloride + bicarbonate concentration). It is mainly constituted by plasma albumin.

Topic: Acid-Base Management | Ref: Ch 40, Section: Anion Gap, Page 294

Q50. Answer: C — Lactic acidosis due to tissue hypoxia

Lactic acidosis is the most common cause of metabolic acidosis in anesthesia and is due to tissue hypoxia. It is associated with increased anion gap.

Topic: Acid-Base Management | Ref: Ch 40, Section: Metabolic Acidosis – Increased Anion Gap, Page 295